PRODUCT DOCUMENTATION · USER MANUAL

Mohar Control Room

What the website does today, how an operator uses it, and exactly where the line falls between what is built and what is still planned.

Scope the nine pages of the running application
Method every screenshot captured from the live build
Rule nothing is described as working unless it was observed working
Chain tip at capture #702

01 What Mohar is

The problem, the users, and what the website is for.

Mohar is a control room for examination paper custody. It is a website an operator watches while sealed question-paper packages move from the printing authority to an examination centre and are opened there.

The problem it addresses

Question papers usually leak because somebody who was permitted to hold them passed them on. The difficulty afterwards is not catching the encryption — it is that no trustworthy record exists of who held the package and when. Handwritten custody registers are completed after the fact and say whatever is convenient.

Mohar's answer is a record of custody that is added to and never edited. Each entry is signed by the device that produced it and linked to the entry before it, so removing or altering one is visible rather than silent.

Who uses the control room

USER	WHAT THEY DO HERE
Control-room operator	Watches packages, refusals and unresolved acts across all centres
Centre staff	Run the unlock ceremony on the Ceremony page and manage fingerprint slots
Auditor or investigator	Reads the full event history of one package and re-verifies the chain

What the operator can monitor

- Packages and their custody risk, ranked
- Access decisions, including every refusal and the reasons for it
- Custody keys, their epoch and their validity
- Enrolled signing devices
- Every recorded act, with the evidence behind it
- The integrity of the record itself

> **HOW TO READ THIS DOCUMENT**
>
> Every feature carries a label. BUILT means it was observed working in the running application. PARTLY BUILT means the interface exists and the document says precisely what part works. PLANNED means it is designed but not demonstrated by the website, and it is kept in section 17.

02 The nine pages

Everything reachable from the sidebar, and what each one is for.

PAGE	PURPOSE	STATUS
Overview	Operational summary: counters, centre map, recent refusals	BUILT
Workflow	Every event for one package, in order, with the signed payload behind each	BUILT
Packages	All sealed bundles, ranked by custody risk	BUILT
Ceremony	The unlock ceremony: camera, package details, fingerprint reads	BUILT
Slots	Which person each fingerprint slot belongs to; enrol from the browser	BUILT

PAGE	PURPOSE	STATUS
Activity	Every recorded act with its full evidence	BUILT
Keys	Custody keys: issue, rotate, and test the access engine	BUILT
Devices	Enrolled signing devices; revoke one	BUILT
Integrity	Recompute the chain; Merkle anchors	PARTLY

The sidebar counters

Four sidebar entries carry a number. These are the values shown at the moment of capture; they change as the system is used.

ENTRY	VALUE	WHAT THE INTERFACE ESTABLISHES
Packages	16	The number of packages listed on the Packages page
Activity	54	Shown in the alert style. The Overview card labels the same figure “awaiting a decision”
Keys	8	Shown in the alert style, alongside the Overview refusal counter for key-related refusals
Devices	47	Matches the Devices page header, which reads “47 active · 1 revoked”

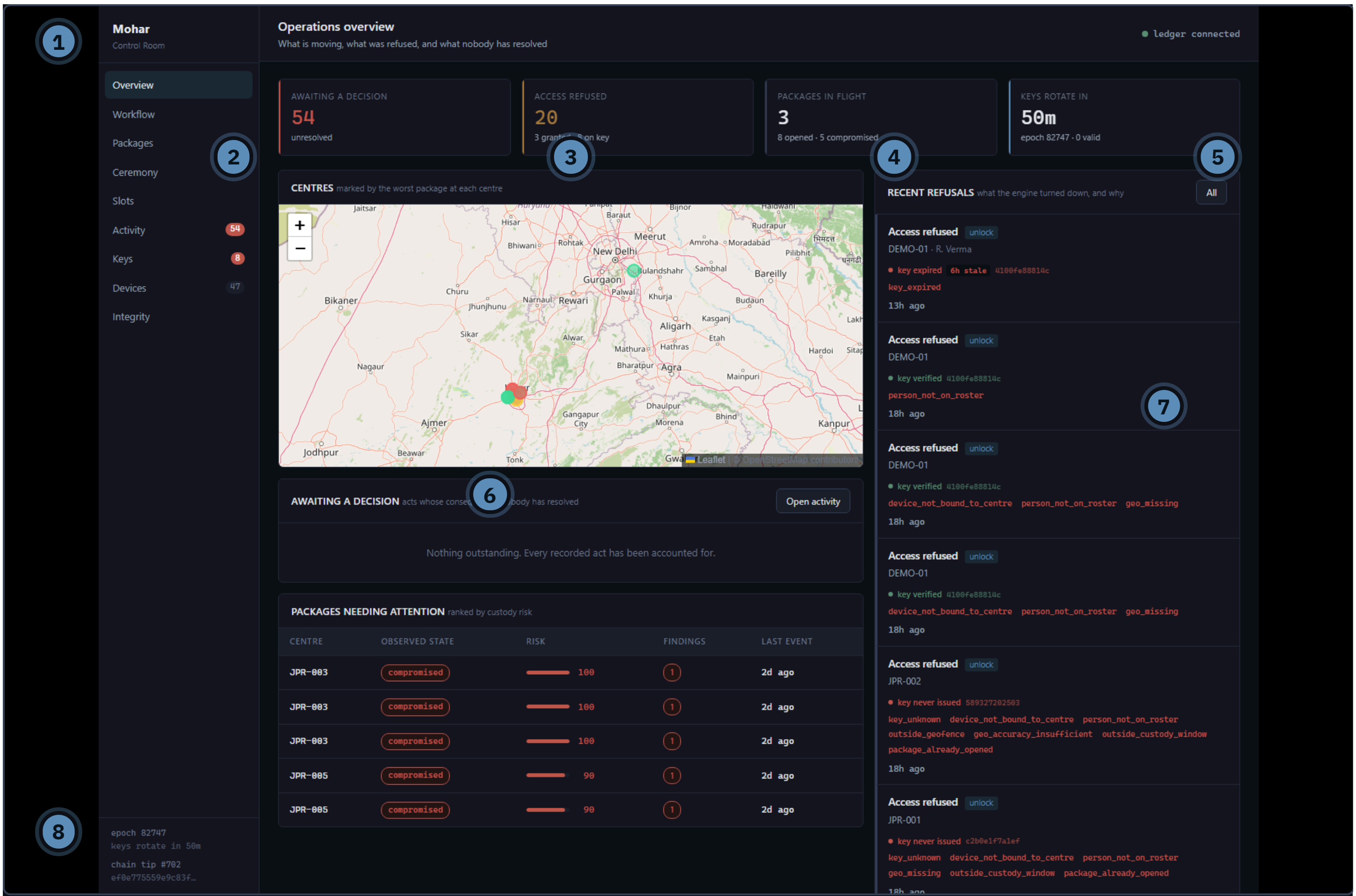
The sidebar footer

Present on every page, so the state of the record is never more than a glance away.

FIELD	AT CAPTURE	MEANING
epoch	82747	The current custody-key epoch
keys rotate in	57m	Time until the next epoch
chain tip	#702	The sequence number of the most recent record
tip hash	ef0e775559e9c83f…	The hash of that record, truncated

03 Overview — the dashboard

The first screen. An operational picture in one view.



Overview, captured from the running build

1. Sidebar navigation with live counters
2. **Awaiting a decision** — unresolved acts
3. **Access refused** — refusals, with a granted comparison
4. **Packages in flight** — package states
5. **Keys rotate in** — countdown and epoch
6. **Centres** map, marked by the worst package at each centre
7. **Recent refusals** — what the engine turned down, and why
8. Footer: epoch, rotation countdown, chain tip and hash

The four cards, exactly as shown

AWAITING A DECISION
54
unresolved

ACCESS REFUSED
20
3 granted · 8 on key

PACKAGES IN FLIGHT
3
8 opened · 5 compromised

KEYS ROTATE IN
57m
epoch 82747 · 0 valid

CARD	WHAT THE INTERFACE COMMUNICATES
Awaiting a decision 54	Fifty-four recorded acts are marked unresolved. The card is the operator's queue: these are acts whose consequences nobody has acknowledged. The same figure appears on the Activity sidebar entry.
Access refused 20	Twenty access attempts were refused. The supporting line gives two further figures shown by the interface: 3 granted and 8 on key — refusals attributable to the custody key, matching the Keys sidebar counter.
Packages in flight 3	Three packages are in flight. The supporting line reports 8 opened and 5 compromised. These states also appear per package on the Packages page.
Keys rotate in 57m	Fifty-seven minutes until the next epoch. epoch 82747 is the current epoch and 0 valid is the count of currently valid keys — the same zero shown on the Keys page as “keys valid now”.

READING “0 VALID”

Zero valid keys is not an error state. The Keys page explains that a key is valid for one six-hour epoch and that expiry is arithmetic on the clock. All 249 keys issued so far belong to earlier epochs, so none is currently valid.

04 Ledger status and the centre map

Two elements present on the Overview screen.

Ledger connection indicator

Top right of every page: a coloured dot and the words ledger connected. It reports whether the website can currently reach the record service. Every screenshot in this document was captured while it read connected.

Centres map

The panel is titled **Centres** and subtitled “marked by the worst package at each centre”. Centres appear as coloured markers on a map of the region, with zoom controls. The marker colour is set by the worst package at that centre, which is what the panel's own subtitle states.

STATED CONSERVATIVELY ON PURPOSE

The interface does not print a colour key, so this document does not assign fixed meanings to individual colours beyond the panel's own description. The per-package state and risk score are read from the Packages page, which does label them.

```
Control Room → Centre map → marker → worst package at that centre
                                   ↓
                     Packages page → that package's state and risk
```

05 Recent refusals

The panel that makes the system's refusals legible.

On the right of the Overview screen, under the heading **Recent refusals** with the subtitle “what the engine turned down, and why”. An **All** button in the corner opens the Activity page.

What one refusal record shows

ELEMENT	EXAMPLE FROM THE CAPTURE
Act	Access refused
Stage label	unlock
Centre, and person if attributed	DEMO-01 · R. Verma
Key badge	key expired · 6h stale and the key fingerprint
Refusal reasons	key_expired
Relative time	13h ago

Example 1 — an expired key

DEMO-01 · R. Verma, stage unlock, key badge key expired · 6h stale, reason key_expired, 13h ago. The operator can read that a key was presented, that it was recognised, and that it had passed its epoch. The person is named, so the refusal is attributable.

Example 2 — a verified key, but the person failed

DEMO-01, key badge key verified, reason person_not_on_roster, 17h ago. The two together are the useful part: the key was accepted and the request still failed, so the problem lies with who presented it rather than with what they presented.

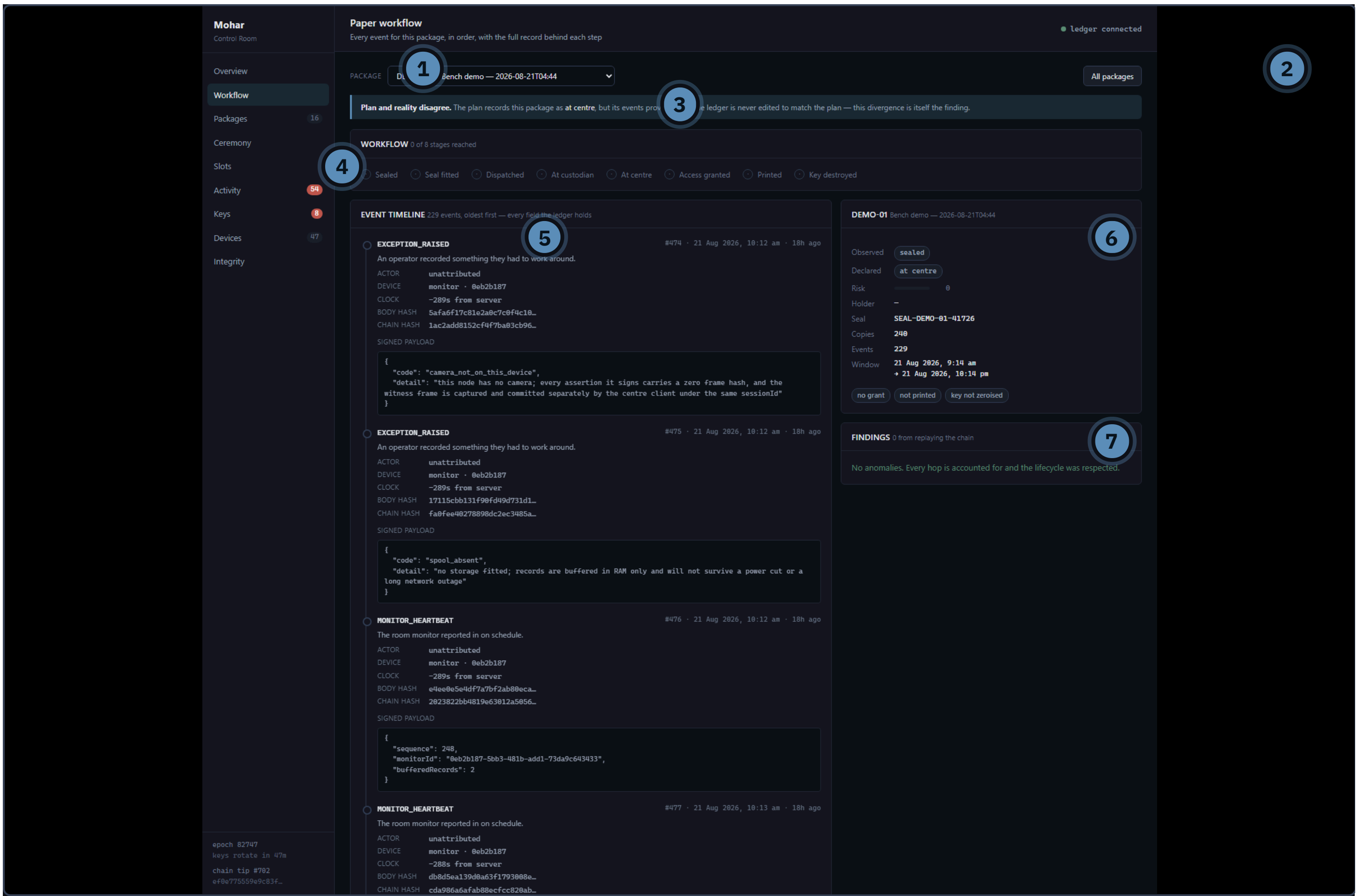
Example 3 — two reasons at once

A record showing both device_not_bound_to_centre and person_not_on_roster. Reasons are listed as separate chips, so an attempt that failed several ways is visibly different from one that failed on a single point.

> CORRECTION — “UNLOCK” IS NOT A BUTTON
>
> The small unlock tag beside “Access refused” is the **custody stage** the attempt was made at — the sixth stage of eight, listed on the Workflow page as “Access granted”. It is a label, not a control.
>
> In the current build the refusal rows are not clickable and carry no action. There is no unlock action in this panel, and this document does not describe one. The only control in the panel is the **All** button, which opens the Activity page.

06 Workflow

Every event for one package, in order.



Workflow with a package selected

1. Package selector — the page is empty until one is chosen
2. **All packages** returns to the Packages list
3. Divergence banner, shown when plan and events disagree
4. **Workflow** — the eight custody stages and how many are reached
5. **Event timeline** — every event, oldest first
6. Package summary: observed, declared, risk, seal, copies, events, window
7. **Findings** — anomalies from replaying the chain

Screen → action → result

SCREEN	OPERATOR ACTION	DISPLAYED RESULT
Workflow, empty	Choose a package from the selector	Timeline, stage tracker and summary load for that package
Workflow, loaded	Read the stage tracker	“0 of 8 stages reached” with each stage listed
Workflow, loaded	Read down the timeline	Each event with kind, sequence, time, actor, device, clock offset, body hash, chain hash and the signed payload

The eight stages shown

Sealed · Seal fitted · Dispatched · At custodian · At centre · Access granted · Printed · Key destroyed

WHAT THE DIVERGENCE BANNER DOES

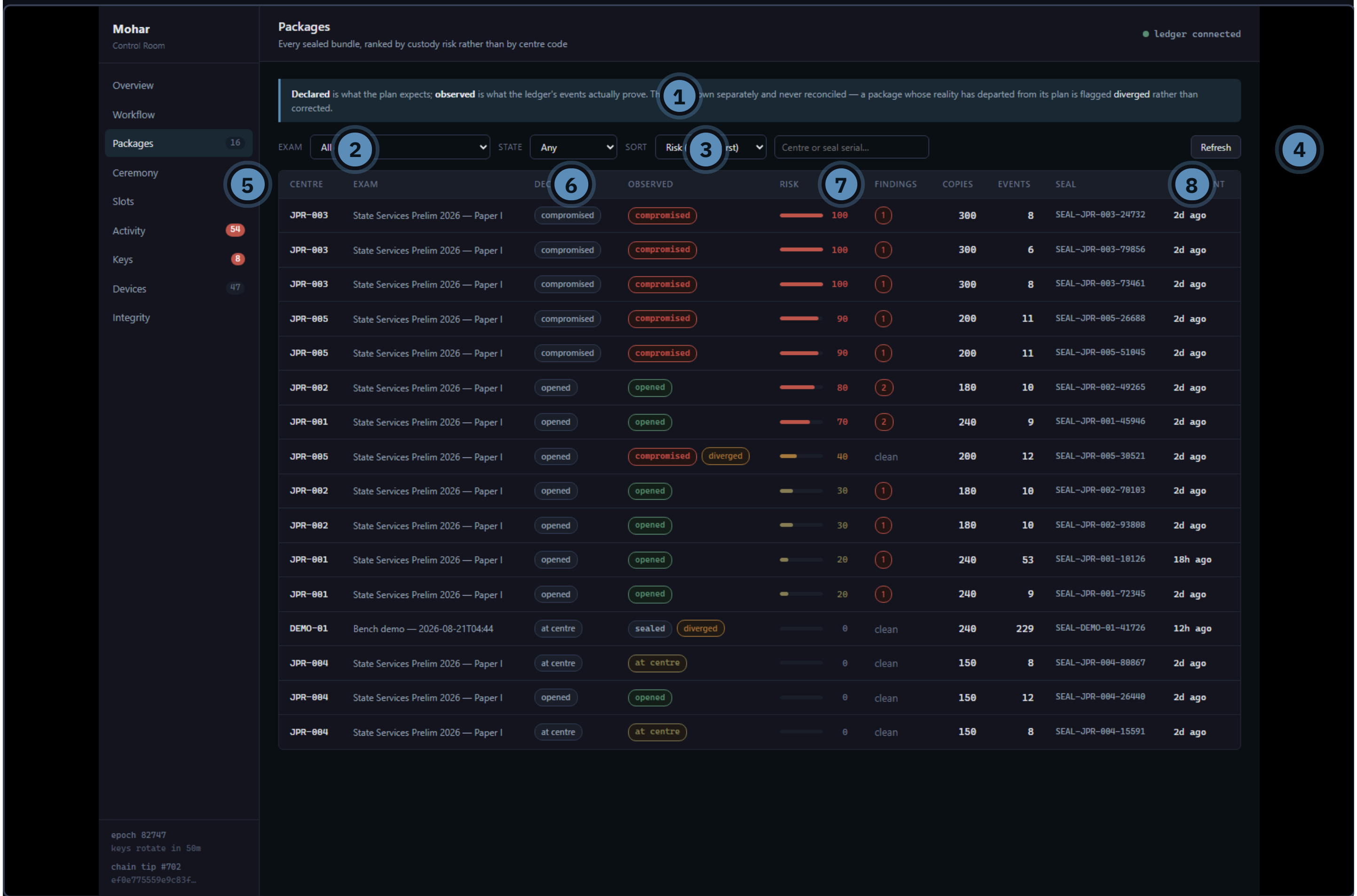
In the capture it reads: the plan records this package as **at centre**, but its events prove **sealed**. The banner states that the record is never edited to match the plan, and that the divergence is itself the finding. Plan and observed state are shown side by side and left to disagree.

THE TIMELINE CARRIES THE ACTUAL EVIDENCE

Each entry shows the body hash, the chain hash and the exact signed payload as JSON. An auditor is not being shown a summary of the record; they are being shown the record.

07 Packages

Every sealed bundle, ranked by custody risk rather than by centre code.



Packages, sixteen rows at capture

1. Explanatory banner: declared versus observed
2. Filters — exam and state
3. Sort, defaulting to highest risk first
4. Search by centre or seal serial, and **Refresh**
5. Centre code and exam name
6. **Declared** and **Observed** state, shown separately
7. **Risk** bar and score, and **Findings** count
8. Copies, events, seal serial and time since the last event

Columns

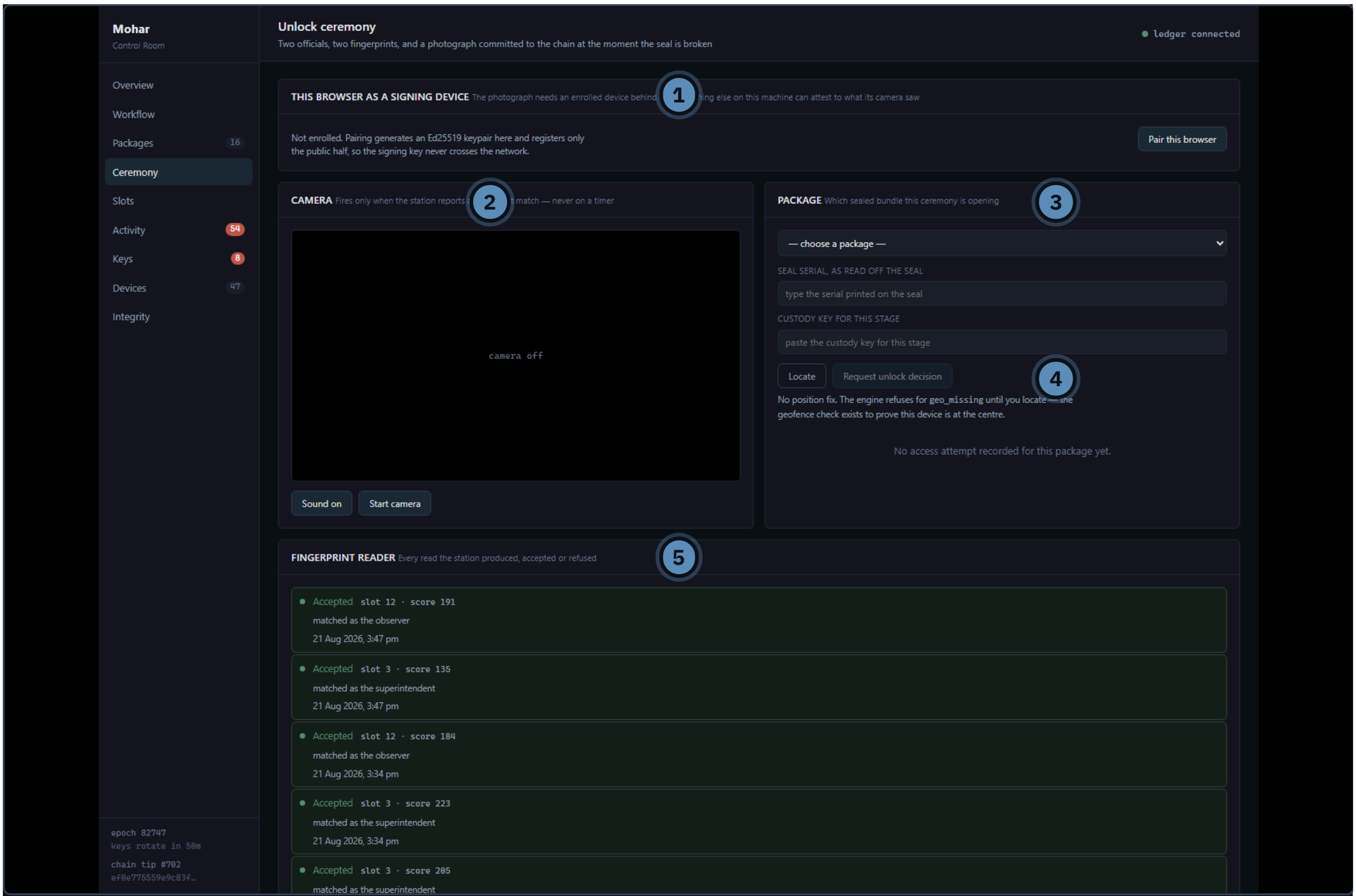
COLUMN	SHOWS
Centre	Centre code, for example JPR-003, DEMO-01
Exam	Exam name
Declared	The state the plan expects
Observed	The state the events prove; a diverged tag appears when the two differ
Risk	A bar and a score from 0 to 100
Findings	A count of findings, or clean
Copies	Number of copies
Events	Number of recorded events for that package
Seal	Seal serial
Last event	Time since the most recent event

DECLARED AND OBSERVED ARE NEVER RECONCILED

The banner on this page states it directly: declared is what the plan expects, observed is what the events prove, and a package whose reality has departed from its plan is flagged **diverged** rather than corrected.

08 Ceremony

The unlock ceremony. Two officials, two fingerprints, and a photograph.



Ceremony, before the browser is paired and before the camera is started

1. Browser pairing panel — the photograph needs an enrolled device behind it
2. **Camera** panel, with **Sound on** and **Start camera**
3. **Package** panel: package selector, seal serial, custody key
4. **Locate** and **Request unlock decision**, and the decision result
5. **Fingerprint reader** — every read the station produced

What the operator must check

1. The browser is paired. Unpaired, the panel offers **Pair this browser** and states that the private key is generated locally and only the public half registered.
2. The camera is started. The panel states that it fires only when the station reports a fingerprint match, never on a timer.
3. The correct package is selected, with its seal serial and custody key entered.

The fingerprint reader panel

Each read appears as its own row, accepted or refused, with the slot number, the match score, the capacity it matched as, and the time. In the capture:

RESULT	SLOT	SCORE	MATCHED AS	TIME
Accepted	12	191	observer	3:47 pm
Accepted	3	135	superintendent	3:47 pm
Accepted	12	184	observer	3:34 pm
Accepted	3	223	superintendent	3:34 pm

OBSERVED WORKING

A complete ceremony was run against this build. Two different fingerprint slots were read, two photographs were captured and their hashes committed, the two-person window closed as confirmed, and the access engine returned **granted** with 19 of 21 checks passed. The two that did not pass reported themselves as not evaluated rather than passing quietly.

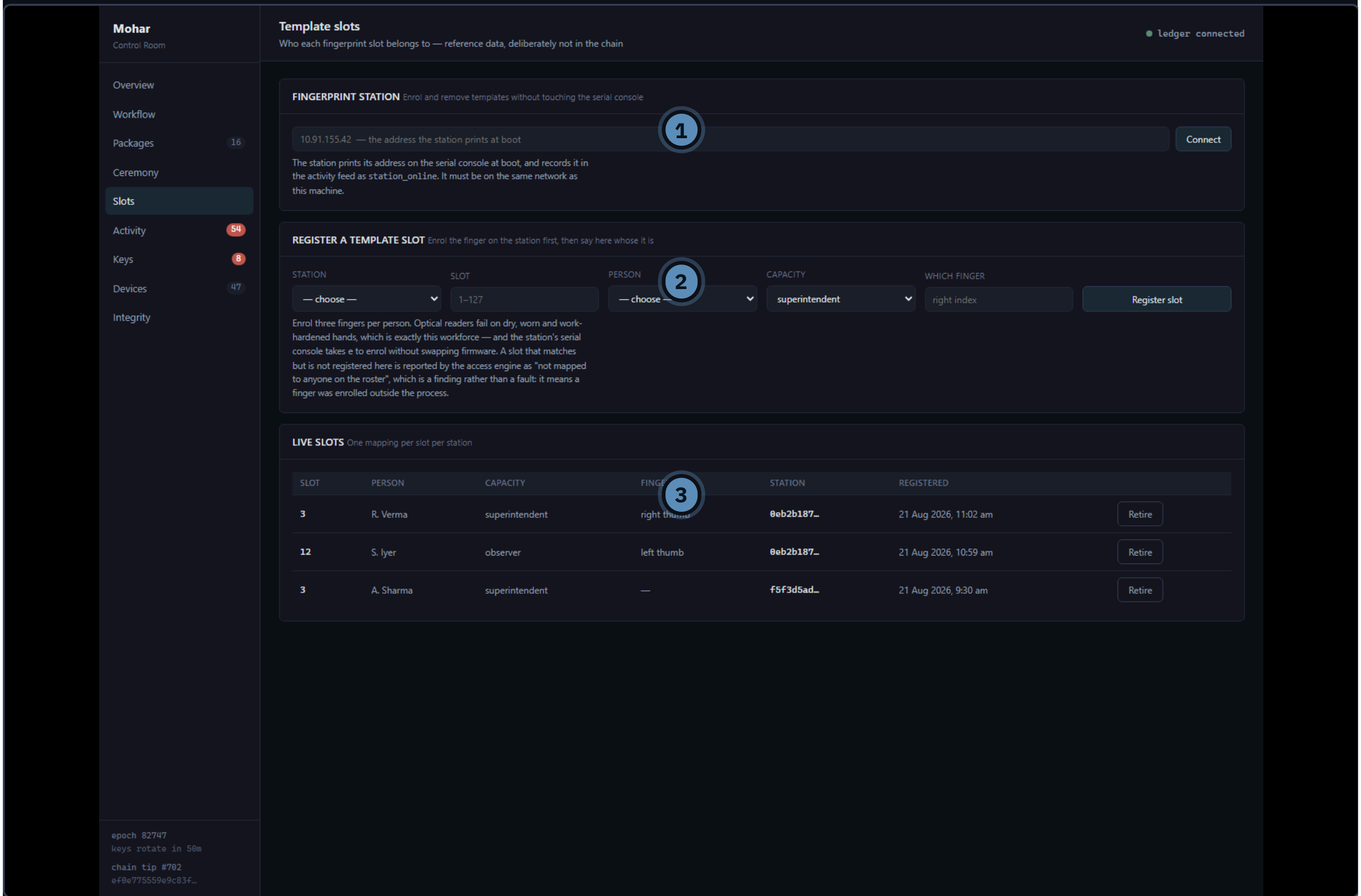
WHERE THE HARDWARE LINE FALLS

The fingerprint reads shown on this page come from a physical ESP32 station with an R307 reader, over the local network. That part is real and was demonstrated.

What the website does **not** do is open anything physically. There is no electronic lock in this build. A granted decision is a recorded decision; the packet is still opened by hand.

09 Slots

Who each fingerprint slot belongs to — reference data, deliberately not in the chain.



Slots, with two live mappings and one from a replaced station

1. **Fingerprint station** — address field and **Connect**
2. **Register a template slot** — station, slot, person, capacity, which finger
3. **Live slots** — one mapping per slot per station, each with **Retire**

The live slots table

SLOT	PERSON	CAPACITY	FINGER	STATION	REGISTERED
3	R. Verma	superintendent	right thumb	0eb2b187…	21 Aug 2026, 11:02 am

SLOT	PERSON	CAPACITY	FINGER	STATION	REGISTERED
12	S. Iyer	observer	left thumb	`0eb2b187…`	21 Aug 2026, 10:59 am
3	A. Sharma	superintendent	—	`f5f3d5ad…`	21 Aug 2026, 9:30 am

Two slots may carry the same number when they belong to different stations, as the third row shows. The mapping is per station, not global.

WHY THIS IS NOT IN THE CHAIN

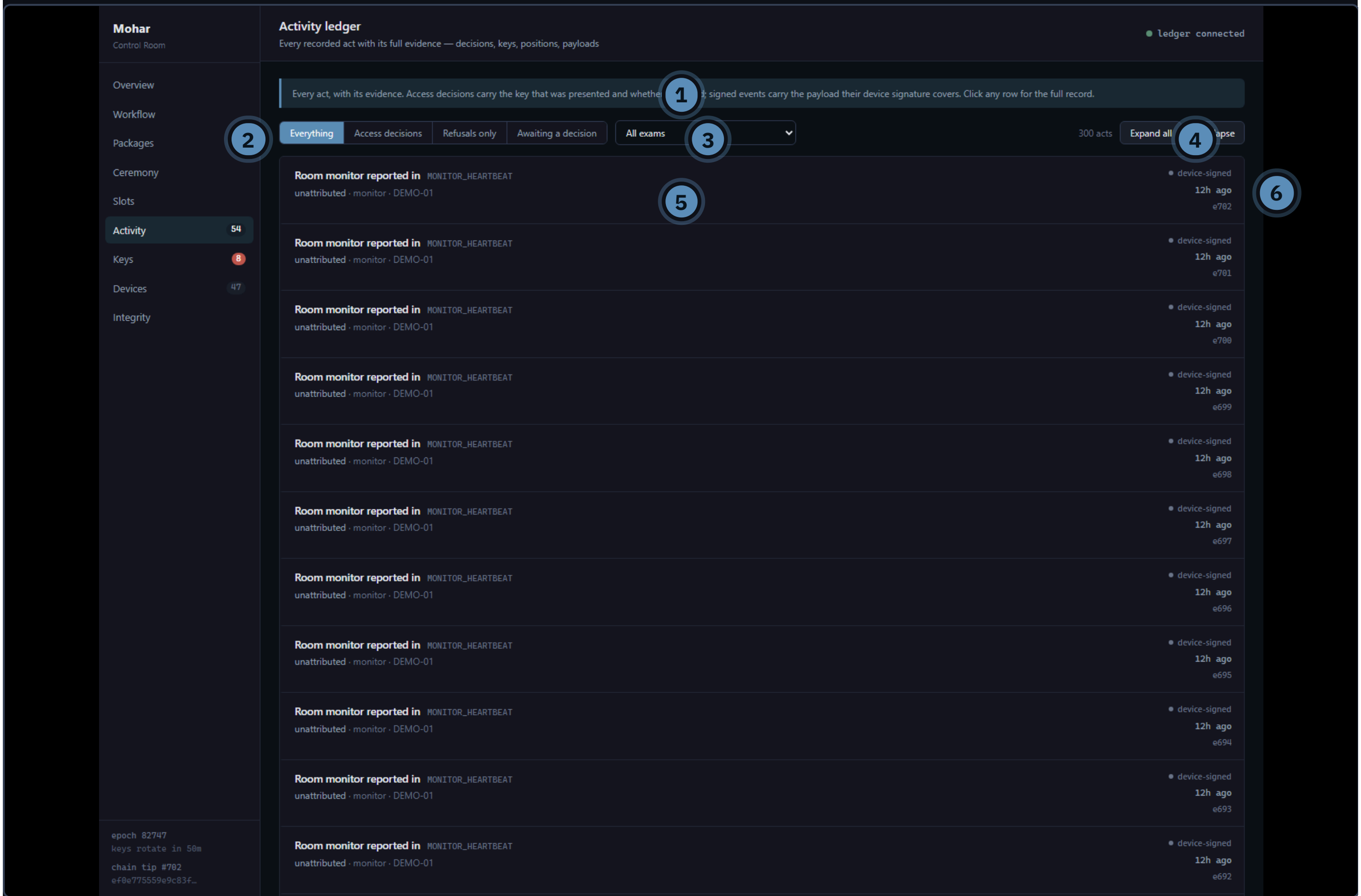
The page subtitle says it: reference data, deliberately not in the chain. The record holds “slot 3 matched, score 187”. Who slot 3 is can be corrected when somebody writes it down wrongly; a signed record cannot. Keeping the two apart is what allows the correction without touching the evidence.

ENROLLING FROM THE BROWSER

With a station connected, the panel enrols and removes fingerprint templates over the network. This was demonstrated: slots 3 and 12 were enrolled from this page, with the browser showing each instruction in turn — place the finger, lift, place again, stored.

10 Activity

Every recorded act with its full evidence.



Activity, showing 300 acts at capture

1. Banner: every act with its evidence; rows can be opened for the full record
2. Filter tabs — Everything, Access decisions, Refusals only, Awaiting a decision
3. Exam filter

4. Act count, **Expand all** and **Collapse**
5. One act: title, event kind, actor, device kind, centre
6. Signature status, relative time, and the event reference

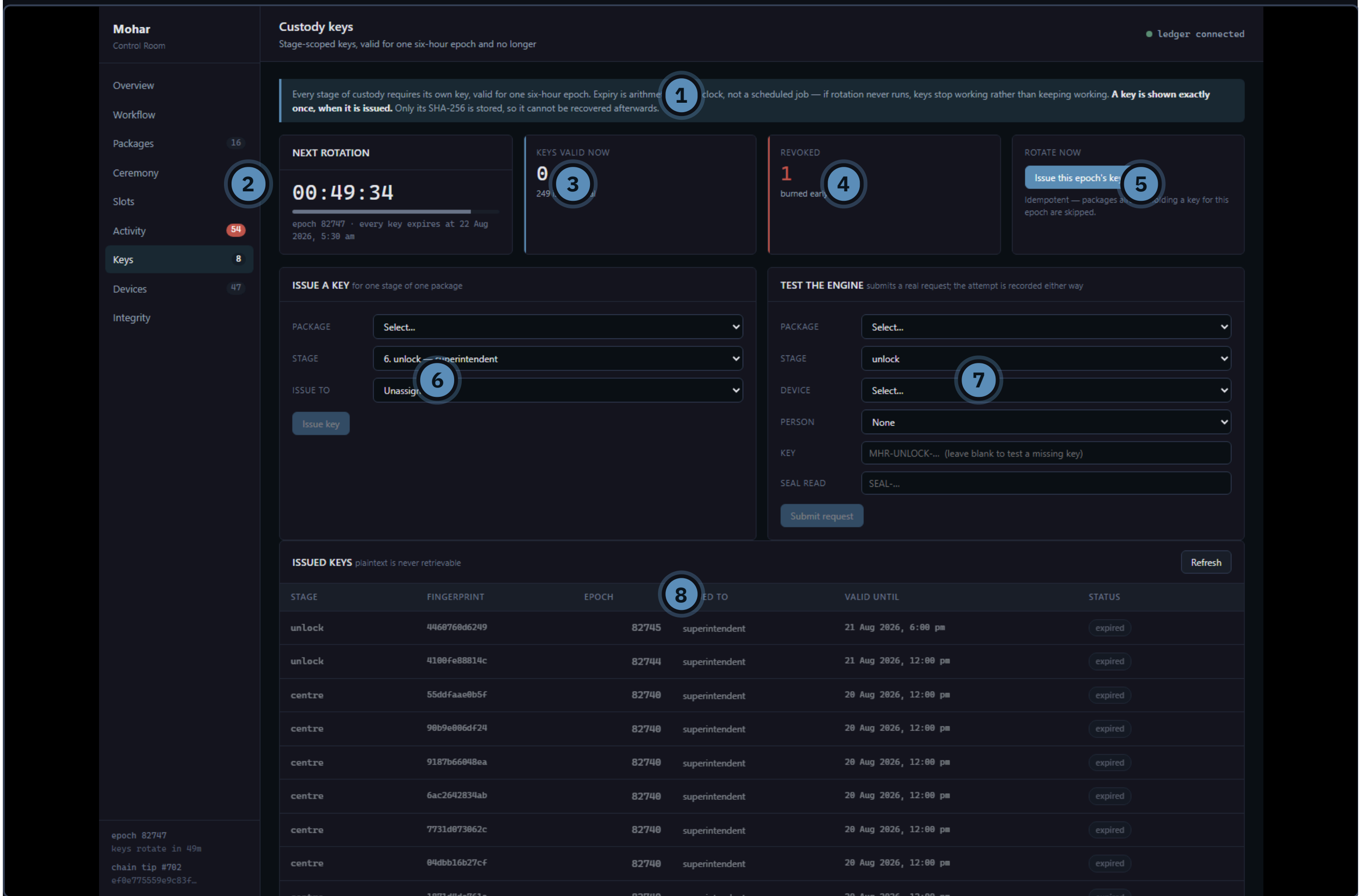
What each row establishes

QUESTION	ANSWERED BY
What happened	The act title, in words, plus the event kind, for example MONITOR_HEARTBEAT
Who did it	The actor, or unattributed where no person is named
On what	Device kind and centre code
When	Relative time, with the exact time available
Is it authentic	device-signed
Where in the record	The event reference, for example e702

> **THE FOUR FILTERS ARE THE OPERATOR'S WORKFLOW**
>
> **Refusals only** narrows to what the engine turned down. **Awaiting a decision** narrows to acts nobody has resolved — the same set the Overview card counts.

11 Keys

Stage-scoped keys, valid for one six-hour epoch and no longer.



Keys, with the rotation countdown running

1. Banner explaining epoch expiry and that a key is shown exactly once
2. **Next rotation** — live countdown, epoch, and expiry time
3. **Keys valid now** — and the total ever issued
4. **Revoked** — keys burned early
5. **Rotate now** — issue this epoch's keys
6. **Issue a key** — package, stage, issue to
7. **Test the engine** — submits a real request
8. **Issued keys** table

Values at capture

CARD	VALUE	SUPPORTING TEXT
Next rotation	00:49:34	epoch 82747 · every key expires at 22 Aug 2026, 5:30 am
Keys valid now	0	249 issued total
Revoked	1	burned early

The countdown corresponds to the Overview card, which read 57m at its own moment of capture, and to the sidebar footer on every page.

The issued keys table

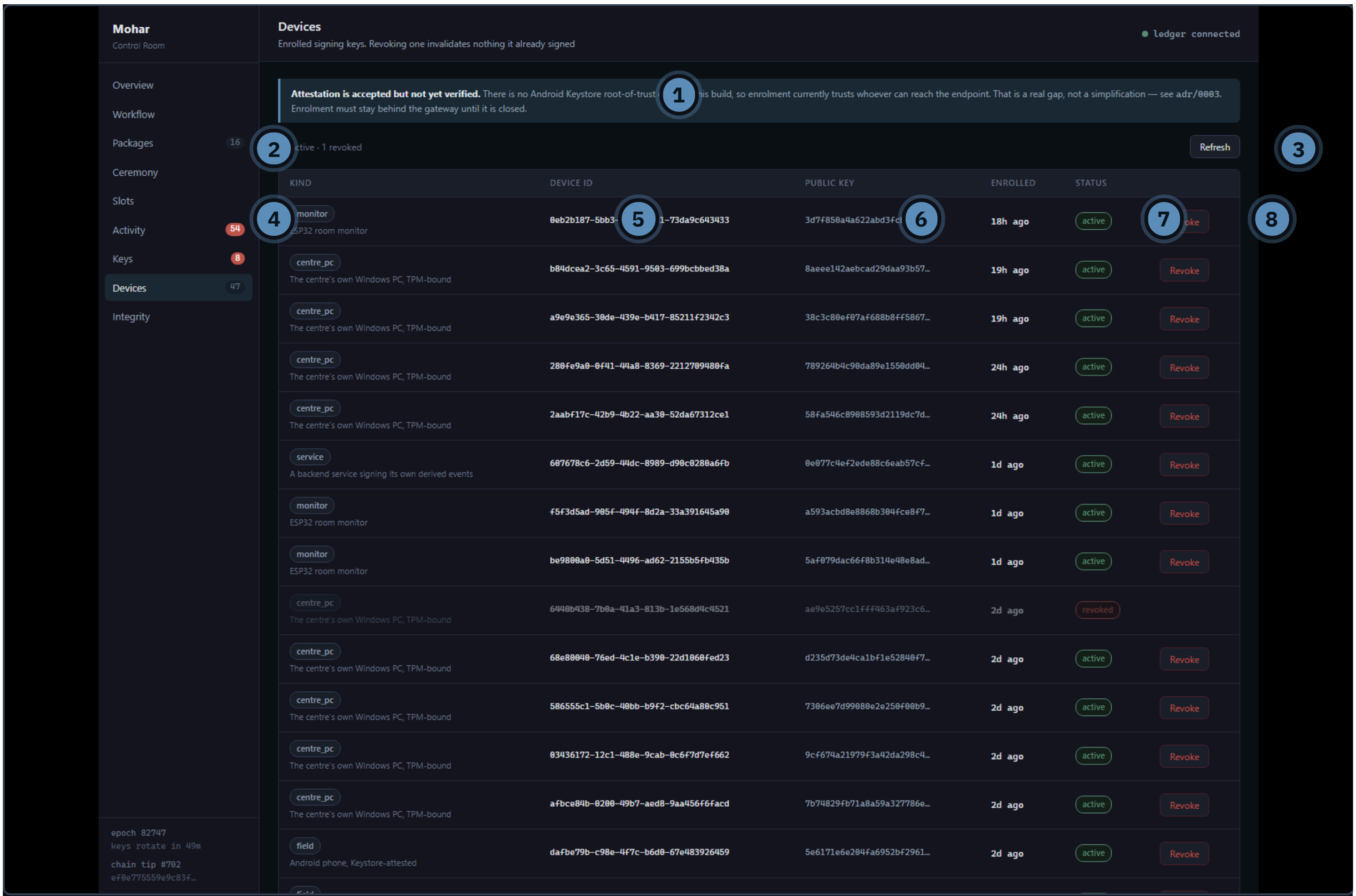
Columns: stage, fingerprint, epoch, issued to, valid until, status. Every row in the capture carries the status expired, which is consistent with “keys valid now: 0” — all of them belong to earlier epochs.

> **TEST THE ENGINE**
>
> This panel submits a genuine access request — package, stage, device, person, key and seal — and its own subtitle notes that the attempt is recorded either way. An operator can reproduce a refusal deliberately, and the refusal is written into the record like any other.

> **A KEY IS SHOWN EXACTLY ONCE**
>
> The banner states that only the SHA-256 of a key is stored and the key cannot be recovered afterwards. This was observed: a key that had expired could not be retrieved, only reissued.

12 Devices

Enrolled signing keys. Revoking one invalidates nothing it already signed.



Devices, 47 active and 1 revoked at capture

1. Banner: attestation is accepted but not yet verified
2. Counts — 47 active, 1 revoked
3. **Refresh**
4. **Kind** — monitor, centre_pc, service, field, each with a description
5. **Device ID**
6. **Public key**, truncated
7. **Enrolled** — how long ago
8. **Status** — active or revoked, with a **Revoke** action

Device kinds shown

KIND	DESCRIPTION IN THE INTERFACE
monitor	ESP32 room monitor
centre_pc	The centre's own Windows PC, TPM-bound
service	A backend service signing its own derived events
field	Android phone, Keystore-attested

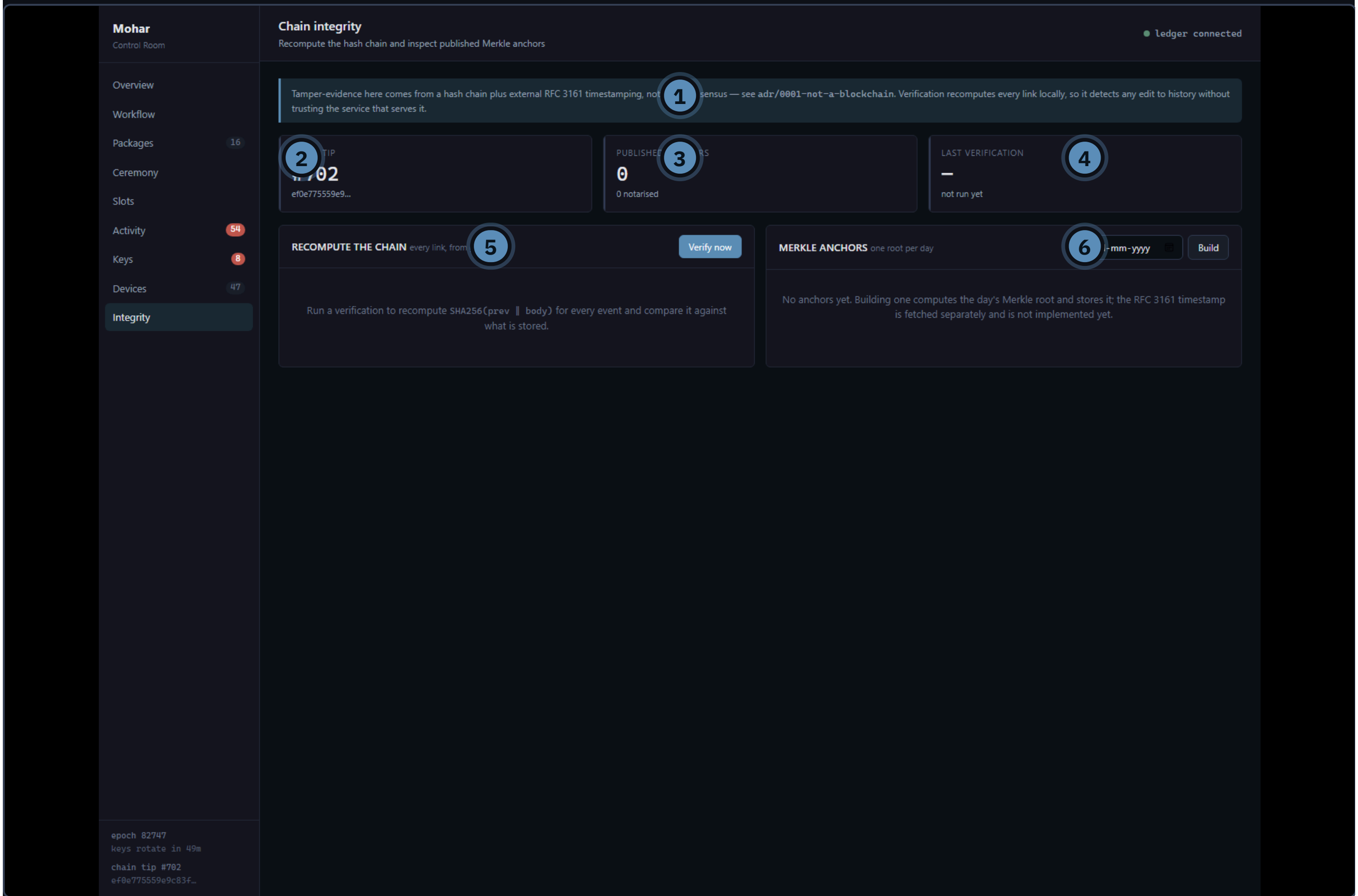
Connecting this to a refusal

The refusal reason device_not_bound_to_centre means the device that made the request is enrolled, but against a different centre from the package it asked about. This page is where that binding is inspected. It was observed in practice: a request made from a device belonging to another centre was refused for exactly this reason, and the refusal cleared once the request came from the station bound to the right centre.

> **THE PAGE STATES ITS OWN GAP**
>
> The banner reads: attestation is accepted but not yet verified; there is no Android Keystore root-of-trust check in this build, so enrolment currently trusts whoever can reach the endpoint. It calls this a real gap rather than a simplification. The centre_pc rows describe themselves as TPM-bound, which is the design; the banner is the authority on what is actually verified today.

13 Integrity

Recompute the hash chain and inspect published Merkle anchors.



Integrity, with no verification run yet

1. Banner: tamper-evidence comes from a hash chain plus external timestamping, not consensus
2. **Chain tip** — #702 and its hash
3. **Published anchors** — 0 notarised
4. **Last verification** — not run yet
5. **Recompute the chain** with a **Verify now** button
6. **Merkle anchors** with a date field and **Build**

What each panel offers

PANEL	WHAT IT DOES	STATUS
Recompute the chain	Recomputes SHA256(prev || body) for every event and compares it against what is stored	BUILT
Merkle anchors	Computes the day's Merkle root and stores it	PARTLY

THE PAGE MARKS ITS OWN UNFINISHED PART

The anchors panel states: building one computes the day's Merkle root and stores it; the RFC 3161 timestamp is fetched separately and is not implemented yet. So the root is computed and stored, and the external notarisation is not done. “0 notarised” on the card is consistent with that.

WHY VERIFICATION IS LOCAL

The banner notes that verification recomputes every link locally, so it detects an edit to history without trusting the service that served it. That is the property worth demonstrating in a viva: the check does not depend on the system being honest.

14 The operator workflow

How the pages are actually used together, in the current build.

```
Open the Control Room                no sign-in exists in this build
        ↓
Overview                             counters, map, recent refusals
        ↓
Something needs attention?               awaiting a decision  ·  a refusal  ·  a risky package
        ↓
Activity  or  Packages                  filter to refusals, or sort by risk
        ↓
Workflow                             open that package, read every event in order
        ↓
Read the reason                         key badge  ·  refusal chips  ·  signed payload
        ↓
Act where the cause lies
   ├ key expired         →  Keys      issue a key for this stage
   ├ slot not mapped     →  Slots     register the slot to a person
   ├ device wrong        →  Devices   inspect or revoke
   └ ceremony needed     →  Ceremony  run it, request the decision
        ↓
The record updates                     the attempt is written before the answer returns
        ↓
Integrity                            recompute the chain when it matters
```

The ceremony path, step by step

#	WHERE	ACTION	RESULT SHOWN
1	Slots	Connect to the station	Reader ready, template count
2	Slots	Enrol a finger into a slot	Place, lift, place again, stored
3	Slots	Register the slot to a person	The mapping appears in Live slots
4	Ceremony	Pair the browser, start the camera	Enrolled device id; live preview
5	Ceremony	Select package, enter seal serial and key	Fields accepted
6	At the reader	First official presents a finger	A read appears; a photograph is committed
7	At the reader	Second official, a different finger	Two-person window closes as confirmed
8	Ceremony	— automatic —	The decision is requested and displayed with its check count
9	Activity	Look at the feed	Every step above is present as a signed record

THERE IS NO SIGN-IN

The build has no authentication. Visiting /signin reaches a page that says so and offers a link into the control room. The documentation does not describe a login step, because there is not one.

15 Built, and not built

The distinction, kept clean.

BUILT Working in the website today

FEATURE	EVIDENCE
Nine navigable pages with live counters	All nine captured from the running build
Overview dashboard, centre map, recent refusals	Section 3 and 5
Per-package event timeline with signed payloads	Section 6
Package list with declared versus observed and risk ranking	Section 7
Unlock ceremony: camera capture, photo hash committed, two-person window	Ceremony run end to end; granted with 19 of 21 checks
Fingerprint reads from a physical ESP32 station over the network	Reads visible on the Ceremony page with slot and score
Enrolling and removing fingerprint templates from the browser	Slots 3 and 12 enrolled this way
Slot-to-person register, with retirement kept on the record	Section 9
Activity feed with four filters and full evidence per act	Section 10
Custody key issue, rotation, and a real engine test	Section 11
Device registry with revocation	Section 12

FEATURE	EVIDENCE
Chain recomputation from genesis	Section 13
Access engine with 21 checks, every attempt recorded before the answer	Observed granted and refused, with reasons

PLANNED **Designed, not demonstrated by the website**

FEATURE	CURRENT STATE
Physical unlocking of a package	No electronic lock exists. A granted decision is a record, not a mechanism.
Electronic seal lock	Not built. The engine's check 20 reports “not evaluated” on every unlock rather than passing.
Room monitor sensors	Firmware written; the door, presence and counting sensors were not obtained.
RFC 3161 external timestamping	The Integrity page states it is not implemented yet.
Device attestation verification	The Devices page states attestation is accepted but not verified.
Authentication and the gateway	No sign-in exists; the service must stay on a local network.
TPM-bound browser credential	The Ceremony page states the key sits in browser storage and calls itself a demonstration credential.
Production key custody	The authority's key share is passphrase-wrapped, not held in an HSM.
Deployment at a real examination centre	Not attempted. All data here is seeded or from bench runs.

WORTH NOTICING

Most of the entries in the second table are stated by the website itself, on the page where they matter. That is a deliberate property: a check nobody ran never appears as a check that passed.

16 Explaining Mohar in a viva

A one-minute answer, then the workflow, then the limits.

The one-minute answer

> “Mohar is a control room for examination paper custody. Question papers usually leak because someone who was allowed to hold them passed them on, and the custody register is written up afterwards. So we built a record that is only ever added to, where each entry is signed by the device that made it and linked to the entry before it — deleting a step leaves a visible hole instead of a tidy table.
>
> The website is where an operator watches that record. The Overview page shows packages, refusals, keys and unresolved acts. Workflow shows every event for one package with the signed data behind it. The Ceremony page runs the unlock: two officials present different fingerprints on a physical reader, the laptop camera photographs each one and commits the photo's hash, and the access engine runs twenty-one checks before answering. Every attempt is recorded before the answer is returned, including refusals — so you can see that a key had expired, or that a person was not on the roster.”

Then walk the workflow

1. Overview — something is unresolved, or refused.
2. Activity or Packages — find which one.

3. Workflow — open that package and read the events in order.
4. Read the refusal reason from the chips and the key badge.
5. Act where the cause is: Keys, Slots, Devices or Ceremony.
6. The record updates, and Integrity can recompute the whole chain.

If asked what it cannot do

Say these before they are found.

- Nothing here physically opens a package. The record is the product.
- An optical fingerprint reader can be fooled. It establishes that a body was present, not which body.
- The two-person rule checks two *different* templates. If one person enrolled two of their own fingers under two names, that check passes — the two photographs would show it.
- There is no authentication yet, so the service must stay on a local network.

> THE LINE THAT LANDS
>
> At Hazaribagh the principal was *allowed* to be in that room. No sensor catches that. What the record does is make it impossible for them to claim afterwards that they were never there.
